Deciding the Factorisation Method

Exam code: 0607

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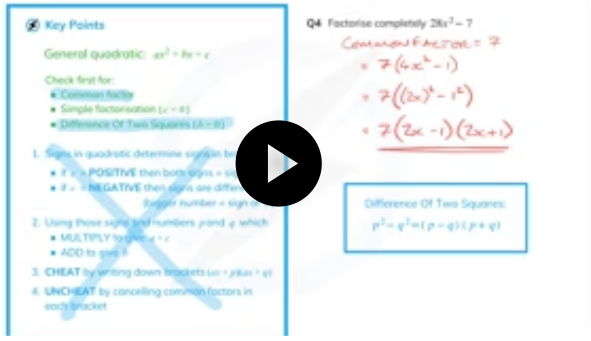

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Quadratics factorising methods

How do I know if an expression factorises?

- The easiest way to check if $ax^2 + bx + c$ factorises is to check if you can find a pair of integers which:
  - Multiply to give ac
  - Sum to give b
  - If you can find integers to satisfy this, the expression must factorise
- There are some alternate methods to check:
  - Method 1: Use a **calculator** to solve the quadratic expression equal to 0
    - Only some calculators have this functionality
    - If the **solutions are integers or fractions** (without square roots), then the quadratic expression will factorise
  - Method 2: Find the value under the square root in the quadratic formula
    - $b^2 - 4ac$
    - If this number is a **square number**, then the quadratic expression will factorise

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it

- Does it have **2 terms only**?
  - Yes, like $x^2 - 7x$
    - Factorise out the highest common factor, *x*
    - $x(x - 7)$
  - Yes, like $x^2 - 9$
    - Use the "difference of two squares" to factorise
    - $(x + 3)(x - 3)$

Does it have **3 terms**?

- Yes, starting with x^2 like $x^2 - 3x - 10$
  - Use "factorising simple quadratics" by finding two numbers that add to -3 and multiply to -10
  - $(x + 2)(x - 5)$
- Yes, starting with ax^2 like $3x^2 + 15x + 18$
  - Check to see if the 3 in front of x^2 is a **common factor** for **all three terms** (which it is in this case), then **factorise it out** of all three terms
  - $3(x^2 + 5x + 6)$
  - The quadratic expression inside the brackets is now x^2 +... , which factorises more easily
  - $3(x + 2)(x + 3)$
- Yes, starting with ax^2 like $3x^2 - 5x - 2$
  - The 3 in front of x^2 is not a common factor for all three term
  - Use "factorising harder quadratics", for example factorising by grouping or factorising using a grid
  - $(3x + 1)(x - 2)$

What other expressions should I be able to factorise?

- You may have a **cubed term** like $x^3 - 3x^2 - 10x$
  - Check to see if *x* is a **common factor** for **all three terms** (which it is in this case), so **factorise it out** of all three terms
  - $x(x^2 - 3x - 10)$
  - The remaining quadratic can then be factorised
  - $x(x + 2)(x - 5)$
- It can also be useful to spot a quadratic in the form $x^2 + 2ax + a^2$
  - This factorises to $(x + a)^2$
  - E.g. $x^2 + 6x + 9 = (x + 3)^2$

Examiner Tips and Tricks

A common **mistake** in the exam is to divide **expressions** by numbers, e.g. $2x^2 + 4x + 2$ becomes $x^2 + 2x + 1$ (which is incorrect).

This can only be done with equations.

Worked Example

Factorise $-8x^2 + 100x - 48$.

Answer:

Spot the common factor of -4 and factorise it out

$$-8x^2 + 100x - 48 = -4(2x^2 - 25x + 12)$$

Check to see if the quadratic in the bracket will factorise using $b^2 - 4ac$

$$\begin{aligned} &(-25)^2 - (4 \times 2 \times 12) \\ &= 625 - 96 \\ &= 529 \end{aligned}$$

529 is a square number (23^2) so the expression will factorise

Factorise $2x^2 - 25x + 12$

We require a pair of numbers which multiply to *ac*, and sum to *b*

$$a \times c = 2 \times 12 = 24$$

The only numbers which multiply to 24 and sum to -25 are

-24 and -1

Split the $-25x$ term into $-24x - x$

$$2x^2 - 24x - x + 12$$

Group and factorise the first two terms, using $2x$ as the common factor
Group and factorise the last two terms using -1 as the common factor

$$2x(x - 12) - 1(x - 12)$$

These factorised terms now have a common factor of $(x - 12)$, so this can be factorised out

$$(2x - 1)(x - 12)$$

Recall that -4 was factorised out at the start

$$-8x^2 + 100x - 48 = -4(2x^2 - 25x + 12) = -4(2x - 1)(x - 12)$$

$$-4(2x - 1)(x - 12)$$

Teacher only

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